

1. Calculate each of the following derivatives.

a) $g(t) = \cot^5(\ln(5t^2))$

b) $h(s) = \cos^4(2s)e^{s^2-3s}$

c) Compute the second derivative $P''(t)$ for the function $P(t) = e^{\arctan(t)}$.

d) $f(x) = \left(\frac{\ln(x^7)}{\arctan(2x)} \right)^9$

2. Show how you can find the derivatives of the following functions *without* the use of the Chain Rule.

a) $g(x) = \frac{x}{2x+1}$

b) $H(x) = \left(\frac{-x^4}{\sqrt[3]{2x}} \right)^7 + \tan\left(2 \sin^{-1}\left(\frac{3}{5} \right) \right)$.

c) $f(x) = \frac{(x+1)^2 e^{x+1}}{3x-2}$

3. Attempt to re-derive the derivative for $y = b^x$.

4. Compute each of the following derivatives.

a) $f(x) = 3^x$

b) $f(x) = 2^{x \cos(x)}$

c) $f(x) = \ln(2)^{\frac{x}{e^{2x}+1}}$

5. Let $f(x) = \begin{cases} x^2 & \text{if } x \leq 2 \\ mx + b & \text{if } x > 2 \end{cases}$. Find values of m and b that make f differentiable everywhere.

3. In algebra we learn about the concepts of *even* and *odd* functions. A function is called *odd* if $f(-x) = -f(x)$ for all values of x . A function is called *even* if $f(-x) = f(x)$ for all values of x .

a) Sketch the graph of a possible odd function. Sketch the graph of a possible even function. *Hint:* It may be helpful to start by plotting a few points on each function and then thinking about what the definitions tell you about other points that must also be on these functions.

b) Show that $y = x^2$ is even. Let n represent any *integer* value. Is $y = x^{2n}$ even for all values of n , odd for all values of n , or does it depend on n ? Justify your answer. What about $y = x^{2n+1}$?

c) Show that the derivative of an *arbitrary* odd function is even and that the derivative of an *arbitrary* even function is odd.

d) Show that if $f(x)$ is odd, then $f(0) = 0$. Is the same true for even functions?

e) Give an example of a function that is both odd and even. Prove that your example is the only function that is both odd and even.